

DTSA 5702

Efficient Programming

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (70.3 / 100)	59.2 / 100
SPECIALIZATION Introduction to High Performance and Parallel Computing		

59

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+61

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course teaches learners the skills needed to develop software to run efficiently in high-performance computing environments or in the cloud. Students will have understand how to find bottlenecks in their programs as well as how to address those bottlenecks. The course will provide a high-level introduction to modern compute node architectures of high-performance and cloud computing instances.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	70.3	100.0	+29.7	MEDIUM
MLOps Engineer	69.1	100.0	+30.9	HIGH
AI Engineer	68.6	100.0	+31.4	HIGH
Applied Scientist	56.2	96.6	+40.4	HIGH
ML Engineer	51.6	81.3	+29.7	MEDIUM
Data Scientist	39.4	100.0	+60.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Model Deployment & Serving 2 of 6 roles

production

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

- 2 **MEDIUM** **Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

- 3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

70.3 → 100.0 (+29.7 pts)

data engineering sets learning tools solutions business time

MLOps Engineer

69.1 → 100.0 (+30.9 pts)

infrastructure data production scientists systems learning machine engineering

AI Engineer

68.6 → 100.0 (+31.4 pts)

data engineering learning infrastructure build technical code systems

Applied Scientist

56.2 → 96.6 (+40.4 pts)

lab literature problems technical data learning systems engineering

ML Engineer

51.6 → 81.3 (+29.7 pts)

data learning machine systems production engineering starting infrastructure

Data Scientist

39.4 → 100.0 (+60.6 pts)

data sets product characteristics presentation evolution scripting different

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.